

CHAPTER - 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

This section identifies the functional dependencies of each entity in the Lenskart E-Commerce Order Management System and analyzes the possible redundancy and data anomalies present in the database design. Functional dependencies describe the relationship between attributes, where one attribute uniquely determines another. Analyzing redundancy helps eliminate duplicate data, while identifying insertion, update, and deletion anomalies ensures data consistency and integrity. This analysis provides the foundation for database normalization and supports the design of an efficient, reliable, and well-structured relational database for the Lenskart E-Commerce Order Management System.

4.2.1 CUSTOMER

Attribute	Description
Customer_ID	Unique customer ID
First_Name	Customer's first name
Last_Name	Customer's last name
Email	Customer email address
Phone_Number	Customer contact number
Password	Customer account password
Gender	Customer gender
Date_of_Birth	Customer date of birth
Address	Customer delivery address
City	Customer city
State	Customer state
Pincode	Customer postal code
Registration_Date	Date of account creation

Functional Dependency

- Customer_ID $\rightarrow$ Customer_Name, Email, Phone_Number, Address

Redundancy Analysis

- Customer details are stored only once using Customer_ID, reducing duplicate customer information.

Data Anomalies

No data anomaly is present because customer details are uniquely stored using **Customer_ID**.

4.2.2 CATEGORY

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Category_Description	Category description

Functional Dependency

- Category_ID → Category_Name

Redundancy Analysis

- Category information is stored only once and referenced by multiple products.

Data Anomalies

No data anomaly is present because category details are uniquely stored using **Category_ID**.

4.2.3 PRODUCT

Functional Dependency

- Product_ID → Product_Name, Brand, Price, Stock, Category_ID

Redundancy Analysis

- Product information is stored only once and linked using Product_ID.

Data Anomalies

No data anomaly is present because each product is uniquely identified by **Product_ID**.

4.2.4 ORDER

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	Not Null
Total_Amount	Not Null
Order_Status	Not Null

Functional Dependency

- $\text{Order_ID} \rightarrow \text{Customer_ID}, \text{Order_Date}, \text{Total_Amount}, \text{Order_Status}$

Redundancy Analysis

- Order details are stored once and referenced by related tables.

Data Anomalies

No data anomaly is present because each order is uniquely identified by **Order_ID**.

4.2.5 ORDER_ITEM

Attribute	Key Type
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Not Null
Price	Not Null

Functional Dependency

- $\text{Order_Item_ID} \rightarrow \text{Order_ID}, \text{Product_ID}, \text{Quantity}, \text{Price}$

Redundancy Analysis

- Each order item is uniquely identified, preventing duplicate product entries.

Data Anomalies

No data anomaly is present because each order item is uniquely identified by **Order_Item_ID**.

4.2.6 PAYMENT

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	Not Null
Payment_Status	Not Null

Functional Dependency

- $\text{Payment_ID} \rightarrow \text{Order_ID}, \text{Payment_Method}, \text{Payment_Status}$

Redundancy Analysis

- Payment details are stored separately to avoid repeating payment information.

Data Anomalies

No data anomaly is present because payment details are uniquely stored using **Payment_ID**.

4.2.7 SHIPMENT

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Tracking_Number	Unique Key
Delivery_Status	Not Null

Functional Dependency

- $\text{Shipment_ID} \rightarrow \text{Order_ID}, \text{Tracking_Number}, \text{Delivery_Status}$

Redundancy Analysis

- Shipment details are maintained separately, avoiding duplicate delivery information.

Data Anomalies

No data anomaly is present because shipment details are uniquely stored using **Shipment_ID**.

4.2.8 ADMIN

Attribute	Key Type
Admin_ID	Primary Key
Admin_Name	Not Null
Email	Unique Key
Password	Not Null

Functional Dependency

- Admin_ID → Admin_Name, Email, Password

Redundancy Analysis

- Administrator details are stored once using Admin_ID.

Data Anomalies

No data anomaly is present because administrator details are uniquely stored using **Admin_ID**.

CONCLUSION

The functional dependency analysis of the Lenskart E-Commerce Order Management System demonstrates that each entity has well-defined relationships between its attributes, ensuring that every non-key attribute is functionally dependent on its primary key. The analysis of redundancy and data anomalies highlights the importance of eliminating duplicate data and preventing insertion, update, and deletion anomalies through proper database design. By identifying these functional dependencies and applying normalization techniques, the database achieves improved data integrity, consistency, and efficiency. This analysis provides a strong foundation for developing a reliable, scalable, and well-structured relational database that

supports the effective management of customers, products, orders, payments, shipments, and administrative operations in the Lenskart E-Commerce Order Management System.